

CSB24084(Assignment 14)

```
import gc
import sys

class Node:
    def __init__(self, name):
        self.name = name
        self.link = None

    def __del__(self):
        print(f"{self.name} is being destroyed")

gc.disable()

print("==== Creating Nodes =====\n")

A = Node("Node A")
B = Node("Node B")

A.link = B
B.link = A

print("Cycle created:")
print("A -> B")
print("B -> A\n")

print("==== Reference Counts =====\n")

print("Reference count of A:", sys.getrefcount(A))
print("Reference count of B:", sys.getrefcount(B))

print("\n==== Deleting References =====\n")

del A
del B

print("Variables A and B deleted")
print("Objects still exist because of cyclic references\n")

print("==== Checking Garbage Collector =====\n")

unreachable_before = len(gc.garbage)

print("Unreachable objects before collection:", unreachable_before)
```

```
print("\n===== Forcing Garbage Collection =====\n")
```

```
collected = gc.collect()
```

```
print("Unreachable objects collected:", collected)
```

```
unreachable_after = len(gc.garbage)
```

```
print("Unreachable objects after collection:", unreachable_after)
```

```
print("\n===== Program Finished =====")
```